

Exam

DIT008: Discrete Mathematics

University of Gothenburg
Lecturer: Jan Gerken

7 January 2026, 8:30 – 12:30

No calculators or other aids are allowed.

You must show your calculations or describe your argument. Unless otherwise stated, if you provide just the final answer, you will not be awarded any points.

The grade boundaries are

- pass (3): at least 20 points (50%)
- pass with credit (4): at least 28 points (70%)
- distinction (5): at least 36 points (90%)

In case of questions, call Jan Gerken at 031-772-14-37.

Good luck!

Problem:	1	2	3	4	5	6	7	8	9	Total
Points:	5	2	4	4	5	5	8	3	4	40

Problem 1 (5 points)

Determine whether the statement forms

$$p \wedge (q \vee r) \quad \text{and} \quad (p \wedge q) \vee (p \wedge r)$$

are logically equivalent. Construct a truth table and include a sentence justifying your answer. Your sentence should show that you understand the meaning of logical equivalence.

Problem 2 (2 points)

Given subsets A and B of a universal set U , provide the definitions for the following in terms of the elements of A and B . No justification needed, 0.5 points each.

- (a) $A \subseteq B$
- (b) $A \cup B$
- (c) $A \cap B$
- (d) $A - B$

Problem 3 (4 points)

Prove the following statement using an element argument and reasoning directly from the definitions.

$$\text{For all sets } A, B \text{ and } C, (A \cup B) \cap C \subseteq A \cup (B \cap C).$$

Problem 4 (4 points)

Prove by induction that

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2 \quad \forall n \geq 1.$$

Problem 5 (5 points)

A game involves placing tokens on a row of n squares. Each square can either be empty, contain one red token, or contain one blue token. However, two adjacent squares cannot both contain blue tokens. Let g_n be the number of different ways to fill a row of n squares following this rule.

- (a) (1 point) Find g_1 and g_2 . Justify your answer.
- (b) (3 points) Find a recurrence relation for g_n where $n \geq 3$. Justify your answer carefully.
- (c) (1 point) Use your recurrence relation to compute g_3 and g_4 .

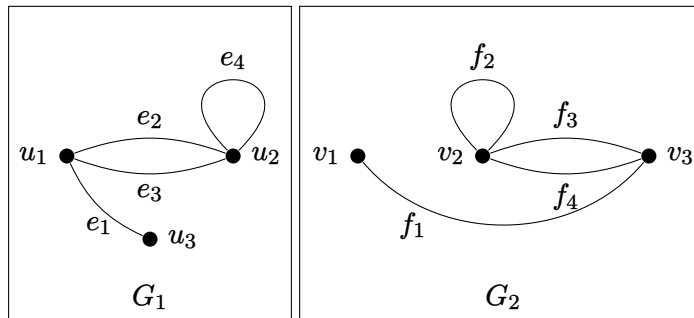
Problem 6 (5 points)

A committee of 8 people consists of 5 teachers and 3 administrators. A subcommittee of 4 people is to be formed.

- (1 point) How many different subcommittees of 4 people can be formed?
- (1 point) How many different subcommittees contain exactly 3 teachers and 1 administrator?
- (1 point) How many different subcommittees contain at least 2 administrators?
- (2 points) If two specific teachers refuse to serve on the same subcommittee (but each is willing to serve if the other does not), how many different subcommittees can be formed?

Problem 7 (8 points)

Consider the following graphs G_1 and G_2



- (2 points) Compute the adjacency matrix A_1 for G_1 with vertex ordering (u_1, u_2, u_3) and the adjacency matrix A_2 for G_2 with vertex ordering (v_1, v_2, v_3) .
- (2 points) G_1 and G_2 are isomorphic. Find the vertex function $g : V(G_1) \rightarrow V(G_2)$ and edge function $h : E(G_1) \rightarrow E(G_2)$ that define the isomorphism.
- (2 points) Let P be the 3×3 matrix defined by

$$P_{ij} = \begin{cases} 1 & \text{if } g(u_i) = v_j \\ 0 & \text{otherwise} \end{cases}.$$

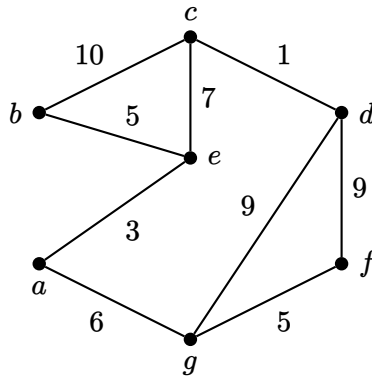
Write down P and $P^\top$, where $P^\top$ is the *transpose* of P with entries $(P^\top)_{ij} = P_{ji}$.

- (2 points) Compute $P^\top A_1 P$. If you did not find A_1 and P above, you may use

$$A_1 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 2 & 0 & 2 \end{pmatrix} \quad P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Problem 8 (3 points)

For the following weighted graph, use Prim's algorithm starting with vertex a to find a minimum spanning tree. Indicate the order in which the edges are added to build the tree.

**Problem 9** (4 points)

Prove that $3x^3 + 5x + 2$ is $\Theta(x^3)$ without using the theorem on polynomial orders.